AI-Powered Credit Risk Intelligence Platform

NeoStats AI Engineer Assignment

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Business Problem

Why Credit Risk Analysis?

Banks need to make fast and accurate lending decisions while maintaining transparency.

Challenges

- Identifying high-risk applicants
- Reducing loan defaults
- Providing explainable decisions
- Supporting business analysts

Solution

- Build a platform that combines:
- Machine Learning
- Explainable AI
- Natural Language Querying
- Interactive Dashboard

System Architecture

Home Credit Dataset

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EDA & Preprocessing

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Machine Learning
(Random Forest)

↓ ──► SHAP Explainability

↓

SQLite Database

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NL → SQL Chatbot

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Streamlit Dashboard

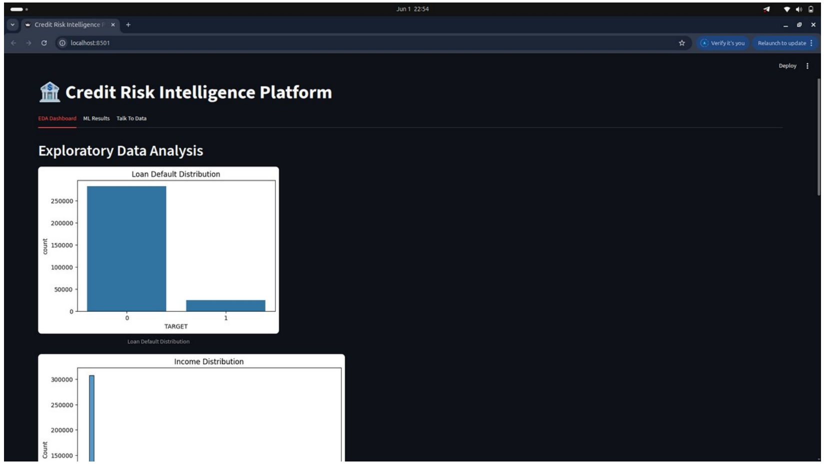
Exploratory Data Analysis

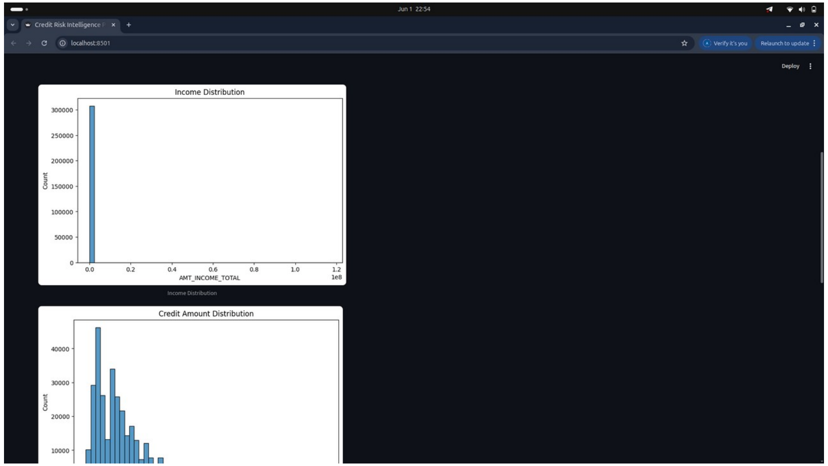
EDA Activities

- Dataset summary
- Feature categorization
- Missing value analysis
- Target distribution analysis
- Correlation analysis

Key Insights

- Default Rate ≈ 8%
- Significant missing values
- Highly imbalanced dataset
- Secondary education is most common





Data Preprocessing

Steps Performed:

- Removed columns with >60% missing values
- Median imputation for numerical features
- Mode imputation for categorical features
- Label encoding

Results:

- Original Dataset : 307,511 × 122
- Processed Dataset : 307,511 × 105

Machine Learning Layer

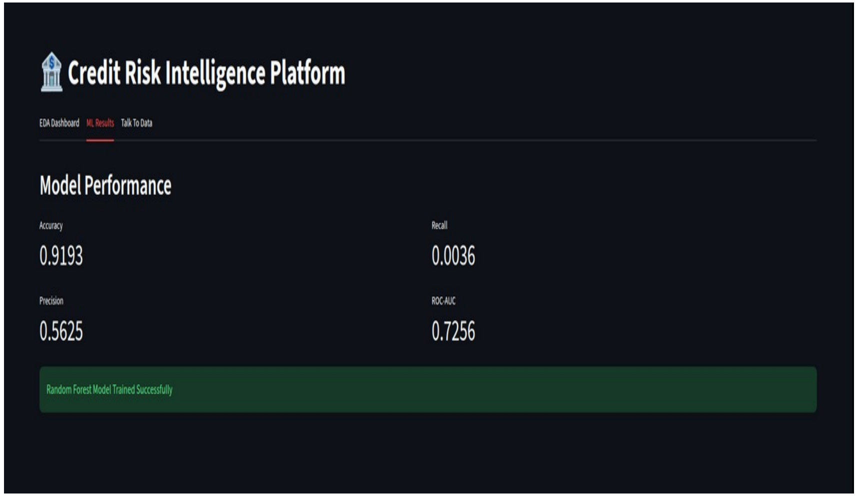
Model used: Random Forest

Why Random Forest?

- Handles tabular data effectively
- Robust to noisy data
- Provides feature importance
- Strong baseline model

Model Performance

Accuracy	91.93%
Precision	56.25%
Recall	0.36%
ROC-AUC	72.56%



Explainable AI

SHAP Explainability

SHAP was used to understand which features influence loan default predictions.

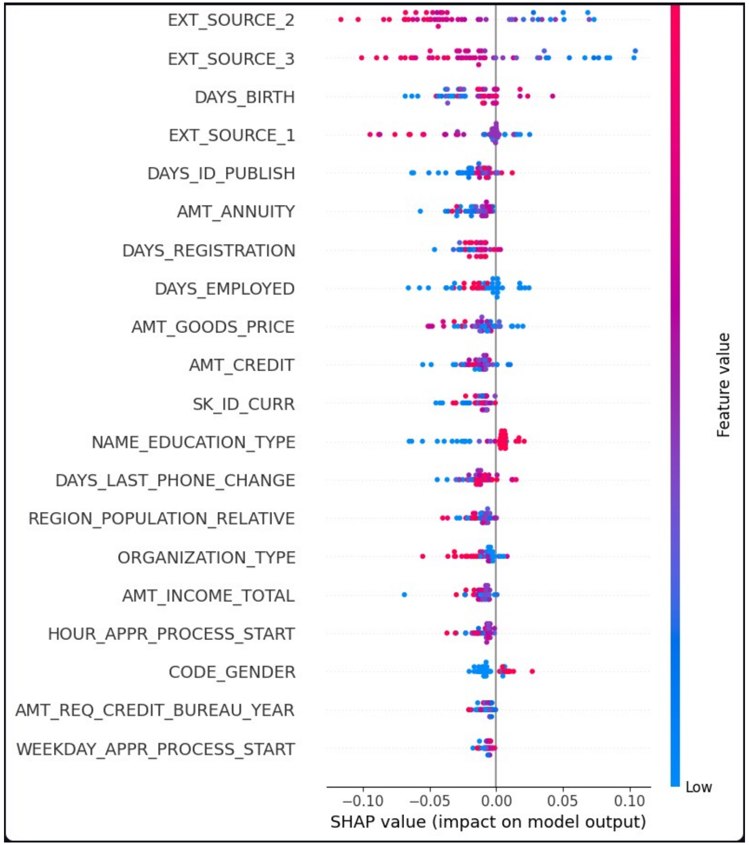
Top Features

- EXT_SOURCE_2
- EXT_SOURCE_3
- DAYS_BIRTH
- EXT_SOURCE_1
- AMT_ANNUITY
- AMT_CREDIT

Benefits

- Transparency
- Trustworthiness
- Regulatory compliance

- If EXT_SOURCE_2 is low and AMT_CREDIT is high → Higher default risk
- Younger applicants with lower external credit scores tend to have higher risk.
- Higher annuity obligations are associated with increased default probability.



Talk To Data (NL → SQL)

Conversational Analytics

- Users can interact with the dataset using natural language.

Example

Question : How many customers defaulted?

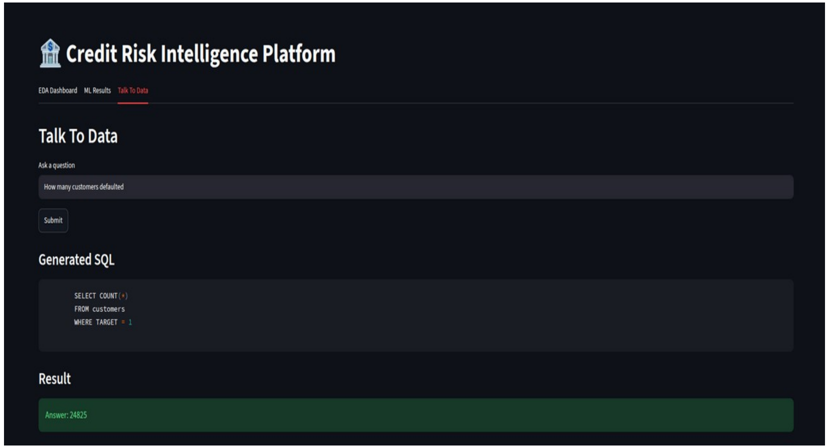
Generated SQL:

```
SELECT COUNT(*)
FROM customers
WHERE TARGET = 1;
```

Result : 24825

Supported Queries

- How many customers defaulted
- Average income
- Average credit amount
- Default rate
- Most common education type



User Interface & Deployment

Streamlit Dashboard

Modules

- EDA Dashboard
- ML Results
- Talk To Data Modules

Deployment

Dockerized using:

Dockerfile

docker-compose.yml

.env.example

Benifits:

- Easy setup
- Portable deployment
- Consistent environment

Conclusion & Future Improvements

Achievements

- Exploratory Data Analysis
- Machine Learning Model
- SHAP Explainability
- NL → SQL Chatbot
- Streamlit Dashboard
- Docker Deployment

Future Enhancements

- XGBoost / LightGBM
- Dynamic LLM-powered SQL generation
- Risk band prediction (Low / Medium / High)
- Cloud deployment
- Real-time APIs

GitHub Repository: https://github.com/Advaid12/credit-risk-intelligence-platform